Supplementary tables for the article "Model-comparison effect size indices for logistic and other generalized linear models: Computation and power analysis"

Simulation results tables

The figures in the main text plot the results of Study 1 (recovery of the whole-model R^2 and of a single focal predictor's eta-squared) and Study 2 (simulation-based required sample size). For Study 1, each Model x N x k combination in the figures is a curve over 10 population effect size values; the tables below summarize accuracy and power collapsed across those population effect size values only (mean absolute bias, RMSE, and mean power gap, by model, sample size, and number of covariates k), keeping the k = 3 vs. k = 5 conditions separate rather than averaging over them, so the full crossed design is preserved – the full per-point results are available in this repository's `simulations/Data/` folder. For Study 2, which has no sample-size or k dimension, the tables report the full underlying data.

Study 1a: accuracy of R^2 (collapsed summary of Figure 1)

Table 1: Accuracy of R2 and adjusted R2, by model, N, and number of covariates k, collapsed across population R2 (data underlying Figure 1).

Model	N	k	Mean |Bias| (R2)	RMSE (R2)	Mean |Bias| (Adjusted R2)	RMSE (Adjusted R2)
Logistic	30	3	0.086	0.155	0.012	0.129
Logistic	30	5	0.144	0.200	0.021	0.140
Logistic	60	3	0.041	0.093	0.004	0.083
Logistic	60	5	0.068	0.112	0.007	0.088
Logistic	90	3	0.027	0.071	0.003	0.065
Logistic	90	5	0.044	0.081	0.003	0.068
Logistic	120	3	0.020	0.059	0.002	0.056
Logistic	120	5	0.032	0.066	0.002	0.057
Multinomial	30	3	0.108	0.147	0.015	0.099
Multinomial	30	5	0.176	0.205	0.023	0.107
Multinomial	60	3	0.053	0.083	0.006	0.064
Multinomial	60	5	0.088	0.112	0.010	0.070
Multinomial	90	3	0.034	0.060	0.003	0.050
Multinomial	90	5	0.057	0.078	0.005	0.053
Multinomial	120	3	0.025	0.049	0.002	0.042
Multinomial	120	5	0.041	0.061	0.003	0.044
Ordinal	30	3	0.061	0.117	0.013	0.100
Ordinal	30	5	0.101	0.149	0.022	0.110
Ordinal	60	3	0.029	0.071	0.005	0.065
Ordinal	60	5	0.046	0.081	0.007	0.067
Ordinal	90	3	0.018	0.054	0.003	0.051
Ordinal	90	5	0.030	0.061	0.004	0.053
Ordinal	120	3	0.014	0.046	0.002	0.044
Ordinal	120	5	0.022	0.050	0.002	0.045
Gaussian	30	3	0.080	0.145	0.007	0.134
Gaussian	30	5	0.136	0.185	0.009	0.149
Gaussian	60	3	0.038	0.093	0.004	0.088

Table 1: Accuracy of R2 and adjusted R2, by model, N, and number of covariates k, collapsed across population R2 (data underlying Figure 1). *(continued)*

Model	N	k	Mean |Bias| (R2)	RMSE (R2)	Mean |Bias| (Adjusted R2)	RMSE (Adjusted R2)
Gaussian	60	5	0.067	0.110	0.005	0.094
Gaussian	90	3	0.026	0.073	0.003	0.070
Gaussian	90	5	0.045	0.083	0.003	0.073
Gaussian	120	3	0.019	0.061	0.002	0.060
Gaussian	120	5	0.033	0.068	0.003	0.061

Study 1a: power of the omnibus test (collapsed summary of Figure 3)

Table 2: Mean |simulated - closed-form| power for R2 and adjusted R2, by model, N, and number of covariates k, collapsed across population R2 (data underlying Figure 3).

Model	N	k	Mean Power Gap (R2)	Mean Power Gap (Adjusted R2)
Logistic	30	3	0.151	0.015
Logistic	30	5	0.235	0.016
Logistic	60	3	0.079	0.010
Logistic	60	5	0.131	0.010
Logistic	90	3	0.051	0.006
Logistic	90	5	0.082	0.007
Logistic	120	3	0.035	0.003
Logistic	120	5	0.059	0.003
Multinomial	30	3	0.238	0.068
Multinomial	30	5	0.315	0.078
Multinomial	60	3	0.118	0.037
Multinomial	60	5	0.163	0.050
Multinomial	90	3	0.070	0.025
Multinomial	90	5	0.099	0.031
Multinomial	120	3	0.049	0.019
Multinomial	120	5	0.070	0.025
Ordinal	30	3	0.107	0.006
Ordinal	30	5	0.164	0.009
Ordinal	60	3	0.049	0.003
Ordinal	60	5	0.078	0.004
Ordinal	90	3	0.029	0.002
Ordinal	90	5	0.048	0.002
Ordinal	120	3	0.018	0.001
Ordinal	120	5	0.033	0.001
Gaussian	30	3	0.218	0.006
Gaussian	30	5	0.332	0.007
Gaussian	60	3	0.126	0.005
Gaussian	60	5	0.206	0.008
Gaussian	90	3	0.083	0.004
Gaussian	90	5	0.130	0.006
Gaussian	120	3	0.057	0.001
Gaussian	120	5	0.096	0.002

Study 1b: accuracy of eta-squared (collapsed summary of Figure 2)

Table 3: Accuracy of eta-squared and adjusted epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 2).

Model	N	k	Mean |Bias| (Eta2)	RMSE (Eta2)	Mean |Bias| (Epsilon2)	RMSE (Epsilon2)
Logistic	30	3	0.032	0.125	0.009	0.120
Logistic	30	5	0.028	0.125	0.009	0.120
Logistic	60	3	0.015	0.082	0.003	0.081
Logistic	60	5	0.016	0.085	0.004	0.083
Logistic	90	3	0.010	0.065	0.002	0.064
Logistic	90	5	0.010	0.066	0.002	0.065
Logistic	120	3	0.007	0.055	0.001	0.055
Logistic	120	5	0.007	0.056	0.001	0.056
Multinomial	30	3	0.034	0.095	0.010	0.088
Multinomial	30	5	0.029	0.092	0.018	0.088
Multinomial	60	3	0.020	0.064	0.005	0.061
Multinomial	60	5	0.019	0.065	0.004	0.062
Multinomial	90	3	0.012	0.050	0.002	0.048
Multinomial	90	5	0.012	0.051	0.003	0.049
Multinomial	120	3	0.008	0.042	0.001	0.041
Multinomial	120	5	0.009	0.043	0.002	0.042
Ordinal	30	3	0.026	0.100	0.010	0.097
Ordinal	30	5	0.025	0.104	0.010	0.100
Ordinal	60	3	0.012	0.065	0.004	0.064
Ordinal	60	5	0.012	0.067	0.005	0.065
Ordinal	90	3	0.008	0.051	0.003	0.051
Ordinal	90	5	0.009	0.053	0.004	0.052
Ordinal	120	3	0.006	0.044	0.002	0.043
Ordinal	120	5	0.006	0.045	0.002	0.044
Gaussian	30	3	0.013	0.112	0.017	0.112
Gaussian	30	5	0.016	0.109	0.027	0.111
Gaussian	60	3	0.007	0.080	0.009	0.081
Gaussian	60	5	0.008	0.078	0.014	0.080
Gaussian	90	3	0.004	0.065	0.006	0.066
Gaussian	90	5	0.006	0.064	0.010	0.065
Gaussian	120	3	0.003	0.056	0.005	0.057
Gaussian	120	5	0.004	0.056	0.007	0.056

Study 1b: power of the single-predictor test (collapsed summary of Figure 4)

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 4).

Model	N	k	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Logistic	30	3	0.058	0.016
Logistic	30	5	0.064	0.018
Logistic	60	3	0.027	0.005
Logistic	60	5	0.032	0.008
Logistic	90	3	0.016	0.003
Logistic	90	5	0.017	0.004
Logistic	120	3	0.011	0.002
Logistic	120	5	0.012	0.001
Multinomial	30	3	0.069	0.019

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 4). *(continued)*

Model	N	k	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Multinomial	30	5	0.074	0.020
Multinomial	60	3	0.032	0.006
Multinomial	60	5	0.034	0.005
Multinomial	90	3	0.020	0.002
Multinomial	90	5	0.019	0.001
Multinomial	120	3	0.011	0.001
Multinomial	120	5	0.011	0.001
Ordinal	30	3	0.044	0.011
Ordinal	30	5	0.052	0.019
Ordinal	60	3	0.017	0.002
Ordinal	60	5	0.017	0.003
Ordinal	90	3	0.009	0.001
Ordinal	90	5	0.009	0.001
Ordinal	120	3	0.005	0.001
Ordinal	120	5	0.006	0.001
Gaussian	30	3	0.049	0.033
Gaussian	30	5	0.043	0.063
Gaussian	60	3	0.032	0.009
Gaussian	60	5	0.024	0.020
Gaussian	90	3	0.022	0.004
Gaussian	90	5	0.019	0.008
Gaussian	120	3	0.015	0.002
Gaussian	120	5	0.012	0.006

Study 2: required sample size (data underlying Figure 5)

Table 5: Required sample size for 80% power, omnibus test: simulated vs. closed-form.

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	312	315
Logistic	0.050	156	157
Logistic	0.075	102	105
Logistic	0.100	77	79
Logistic	0.125	60	63
Logistic	0.150	51	52
Logistic	0.175	42	45
Logistic	0.200	38	39
Multinomial	0.025	246	248
Multinomial	0.050	122	124
Multinomial	0.075	80	83
Multinomial	0.100	59	62
Multinomial	0.125	46	50
Multinomial	0.150	38	41
Multinomial	0.175	32	35
Multinomial	0.200	28	31
Ordinal	0.025	198	198
Ordinal	0.050	98	99
Ordinal	0.075	65	66
Ordinal	0.100	49	50
Ordinal	0.125	39	40
Ordinal	0.150	32	33

Table 5: Required sample size for 80% power, omnibus test: simulated vs. closed-form. *(continued)*

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Ordinal	0.175	27	28
Ordinal	0.200	24	25

Table 6: Required sample size for 80% power, single-predictor test: simulated vs. closed-form.

Model	Population Eta2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	226	226
Logistic	0.050	114	113
Logistic	0.075	75	75
Logistic	0.100	56	57
Logistic	0.125	44	45
Logistic	0.150	37	38
Logistic	0.175	32	32
Logistic	0.200	29	28
Multinomial	0.025	176	175
Multinomial	0.050	87	88
Multinomial	0.075	57	58
Multinomial	0.100	43	44
Multinomial	0.125	34	35
Multinomial	0.150	28	29
Multinomial	0.175	25	25
Multinomial	0.200	23	22
Ordinal	0.025	144	143
Ordinal	0.050	73	71
Ordinal	0.075	48	48
Ordinal	0.100	36	36
Ordinal	0.125	29	29
Ordinal	0.150	25	24
Ordinal	0.175	21	20
Ordinal	0.200	19	18